LPK.DE | LPKF Laser & Electronics | Full investment artifact

Full active edition | Financial methods and portfolio Base overrides preserved.

Active approved edition: 30 September 2026. Framework 8.0.

Q3 2027 ceiling: EUR 35.1892 per share, valuing Q3 2027 through Q2 2028. **Analytical Base: 78/100. Protected Excel Base: 78/100.**

Valuation method: 20x operating profit. EUR millions / 24.5M diluted shares. Same adjusted operating-profit bridge; no hypothetical funding dilution added. Financial amounts: EUR M. Price, upside and ranking remain workbook-owned. This update adopts the approved proposal; it is not a new earnings release or a fresh stock-price quote.

Approved breadcrumb inputs and source images

These probabilities are explicit investment judgments under Framework 8.0, not measured statistical frequencies or claims of confirmed orders. Unconfirmed but credible evidence can affect the model when its company-specific transmission is quantified.

Higher equipment capture with delayed substrate adoption

Active probability: 35% joint supplier-selection and delay case.

Treat the 5-8 tools report and unnamed customer LOI as potentially the same business. Strong selection supports capture, while the micro-crack delay moves part of equipment acceptance into 2028.

Operating bridge: One joint conditional case: glass-tool capture 70% ->80%; shift 25% of the existing 2027 glass revenue into the same quarters of 2028. Incremental revenue earns existing glass GM less the existing opex ratio. No second LOI sale added.

Limits and counter-evidence: The source does not give toolASP or recognition date. Capture revision and 25% delay fraction are analyst bounds. Equipment can precede substrate mass production, so only a quarter of 2027 revenue shifts. Current analytical 78 already reflects the earlier update. No new dilution assumed.

Sources: GLASS, LPKF.

Transmission field	Active treatment
Probability	35% joint supplier-selection and delay case
Recognition	One joint conditional case: glass-tool capture 70% ->80%; shift 25% of the existing 2027 glass revenue into the same quarters of 2028. Incremental revenue earns existing glass GM less the existing opex ratio. No second LOI sale added.
Overlap / uncertainty	The source does not give toolASP or recognition date. Capture revision and 25% delay fraction are analyst bounds. Equipment can precede substrate mass production, so only a quarter of 2027 revenue shifts. Current analytical 78 already reflects the earlier update. No new dilution assumed.

Transmission field	Active treatment
Raise probability	Timely qualifying milestones, comparable revenue scope, stronger customer allocation and observed unit economics.
Lower probability	Delay, customer substitution, weaker realized economics, overlap or evidence contradicting the stated source scope.

1. Thesis, products and competitive position

HELD. TEMPORAL bottleneck with patent protection: LIDE creates fine, crack-free through-glass vias needed for advanced glass packaging. Qualification and IP support capture, while industry multi-sourcing prevents an absolute monopoly claim. The investment case is the conversion from development tools to factory equipment, followed by a wider glass-processing portfolio.

Process, qualification and economic capture

LIDE modifies selected paths inside glass with a laser and then chemically etches those paths to form through-glass vias, cavities and microstructures. The original research identifies sidewall quality, aspect ratio, process recipes, defect control and throughput as the sources of differentiation. Its headline 1:50 aspect ratio and sidewall-quality claims are process benchmarks, not evidence that customers require the maximum specification or that competing tools cannot meet production requirements.

The April 20 management statement that more than 80% of major global players selected LPKF equipment for validation describes a broad installed qualification base. Validation, initial orders, factory acceptance and production share are separate milestones. Patent scope and enforcement against alternative implementations also require evidence. The investment case therefore depends on leading-vendor capture of accepted production equipment, with services and the legacy businesses contributing separately.

Three glass categories with different revenue consequences

Category	Role	Treatment in the operating thesis
Glass carrier	Temporary manufacturing support	A CoPoS carrier announcement does not establish a final-package TGV order.
Glass-core substrate	Core of the finished package, potentially requiring dense TGVs	Relevant to LIDE when the customer's process, production timing and equipment allocation are identified.
Glass interposer / optical glass	Different electrical or optical integration architecture	Needs an explicit product roadmap and process wallet; no automatic additional CPO revenue.

The July source notes describe a TSMC CoPoS sequence of validation in 2026, pilot production in 2027 and mass production in the second half of 2028, while putting a subsequent glass-core phase later. These are dated roadmap/channel observations, not a current customer purchase schedule. The distinction remains essential: a panel-packaging ramp can proceed without a final glass-core substrate. The group-level glass forecast below must not be described as a confirmed TSMC-specific allocation.

Production competition and the downstream yield gate

The original competitive map includes Innolux/ITRI/Contrel, 鈦昇, F&S Electron at Absolics, JNTC's internally developed equipment and photosensitive-glass routes. Published via dimensions, aspect ratios and throughput are not directly comparable without common yield and cost conditions. The useful test is whether an alternative qualifies at a customer's required density, defect rate and throughput. JNTC's vertical model also illustrates that growth in substrate volume need not create equivalent merchant equipment spending.

LPKF's formation step connects to SCHMID wet processing and MKS/Atotech adhesion and plating chemistry. Copper fill, uniformity, microcracks, inspection and panel-level process capability remain system constraints that a laser tool alone cannot resolve. The original research's CpK 1.33 reference is a process-control checkpoint, not a disclosed group revenue constraint. Taiwanese wet-process suppliers 弘塑 and 辛耘 are relevant competitive checks. A good via demonstration is therefore insufficient evidence of a qualified, high-yield package production line.

Ecosystem and adjacent applications

Samsung Electro-Mechanics, SKC/Absolics, Ibiden, Innolux, DNP and LG Innotek are substrate-program checkpoints; their announcements should be tagged by the actual glass category and formation vendor. Intel's installed research capability and possible licensing or outsourcing routes do not establish new direct LPKF orders. Corning and AGC supply upstream glass, while Onto's metrology and the Fraunhofer IZM ecosystem address other process steps. These relationships support the industry map without assigning a purchase order to every participant.

Optical waveguides, electrical vias and PIC coupling in one substrate could increase the number of useful glass-processing steps. That remains an architecture opportunity until production specifications, customers and tool economics are demonstrated. Photonics-SOI can be complementary because the photonic circuit and its package occupy different layers. Robotics, medical and solar applications support a broader technology thesis, but each requires its own recognition bridge. European technology ownership is a strategic context, not a guaranteed acquisition premium or state-funded sales floor.

Research provenance: the product and competitive analysis in this section synthesizes the supplied original research. Dated market-share, process and channel claims retain their original evidence status; this integration is not a new independent verification of those claims. Current earnings and model assumptions are set out below.

2. Earnings, guidance and operating assumptions

Latest actuals and guidance

H1 2026 revenue is €36.5M and adjusted EBIT is -€10.4M; reported EBIT is -€14.1M. The full-year guide remains €105-120M revenue and -3% to 4.5% adjusted EBIT margin. The review uses the upper guided endpoint: €120M revenue and €5.4M adjusted EBIT, requiring €83.5M revenue and €15.8M adjusted EBIT in H2. Q3/Q4 are Model, not company quarterly guidance. [LPK-1]

Current production evidence and model treatment

On September 23, LPKF disclosed a production-equipment letter of intent from an existing Asian advanced-packaging customer that has fully qualified LIDE. Its stated value is in the mid-single-digit millions of euros. The customer is unnamed and the binding order, delivery and acceptance dates are not disclosed. The company also said it is evaluating financing alternatives that may include equity. [LPK-7]

Montega's September 28 research, following management discussions, estimates 3-6 systems and H1 2027 revenue recognition, with potential follow-on orders in the tens of millions of euros in H2 2027. These are attributed analyst expectations, not company guidance, reported sales or an independently verified second customer. Its share-price target is not a valuation input. [LPK-8]

The user-supplied September 28 expert check attributes 5-8 LPKF tools to Samsung Electro-Mechanics, a 25,000-sheet monthly capacity objective in 2028 and 60-70% yield. It also describes Nvidia interest in SEMCO, DNP and BOE. Tool prices, customer identities behind the LPKF LOI, gross versus saleable sheet capacity and acceptance dates remain unresolved. Nvidia interest does not establish an LPKF award. [LPK-9]

The claimed microcrack-driven delay remains contested. TrendForce's August report includes a subsequent Samsung response denying failed customer qualification and differentiating H2 2027 glass-core material from 2028 finished substrates. This is a reported company response; the underlying company statement was not independently obtained. Do not collapse these two production stages into one delay or attribute a confirmed failure to LPKF's via process. [LPK-10]

Active treatment: the 35% joint scenario increases glass-tool capture from 70% to 80% conditionally and shifts 25% of reference 2027 glass revenue to corresponding 2028 quarters. The LOI and reported tools may overlap and are not separate orders. Product margins, cost ratios, 20x OP and 24.5M shares remain unchanged.

The catalyst score is 5/5 because the next 6-18 months now contain a qualified-customer production commitment and a dated recognition expectation. All other scoring categories and the four existing risk deductions remain in force. This improves the assessed likelihood of the operating thesis; the distinct approved probability bridge changes the operating forecast and ceiling without another score change.

TAM definition and recognition

Management defines the €1.7B 2030 TAM as equipment needed if all relevant production switched to glass and installed its tools in that year. It is neither a disclosed annual recurring-sales forecast nor LPKF revenue. The model therefore calculates the change in installed equipment need before applying product wallets and capture. The 70% ambition relates to LPKF's competitive objective; it is not a signed allocation. First production ramps are expected in 2027, with high volume in 2029-2030. [LPK-2]

This single ceiling case assumes full-glass equipment pools of €700/900/1,150/1,420/1,700M in 2026-2030 and adopted glass shares of 3%/8%/22%/48%/80%. Intermediate pools and adoption are LOW-confidence model inputs, with 2030 anchored to the company TAM. Incremental annual spend is current adopted pool minus the preceding year's adopted pool. The adopted share measures physical deployment. The separately stated 35% joint capture/timing adjustment feeds the active glass revenue column below.

Year	Full-glass pool	Adopted share	Installed need	Net new pool	Glass revenue	RDL revenue
2027	900.00	8%	72.00	51.00	30.925	2.04
2028	1,150.00	22%	253.00	181.00	122.543	7.24
2029	1,420.00	48%	681.60	428.60	283.519	17.14
2030	1,700.00	80%	1,360.00	678.40	448.762	27.14

Product capture and demand bridge

The reference glass bridge receives 90% of incremental equipment spending at 70% LPKF capture; the active glass sales include the 35%-weighted capture and timing adjustment. Separately, RDL receives 10% at 40% capture. The wallet split and RDL capture are LOW-confidence estimates; RDL has a smaller opportunity and weaker differentiation than LIDE. The reference wallet assumptions imply 67% aggregate capture before the approved glass adjustment. Glass bonding has no separately quantified wallet and is not added again. LPKF-specific optical transmission through glass is a later architecture; the model includes no CPO equipment revenue in the present horizon without a quantified process/order bridge. That does not change AAOI's earlier external-laser timeline. [LPK-2, LPK-3]

Development/SMT/Solar/Welding revenues are separate Plan-B vectors, anchored to the actual segment totals and disclosed direction of demand. Annual inputs are visible in AE:AT. H2 revenue allocation is €7M glass, €1M RDL, €16.9M Development, €30M SMT, €13.6M Solar and €15M Welding. The split within Electronics is an estimate, not company-reported product revenue.

Calibration, margins and operating leverage

The H1 segment anchor is Electronics €12.8M, Development €11.1M, Solar €4.4M and Welding €8.2M, summing exactly to reported revenue. This validates the group anchor, not the inferred LIDE split. LPKF reports expenses by nature; product gross margins are not disclosed. The model assigns independent gross margins and then deducts opex. An H1 45% proxy gross margin produces €26.825M proxy opex to reconcile the actual adjusted EBIT. These are analytical reclassifications, not reported gross profit. [LPK-4]

Forward glass GM is 60% in 2027 and 62% thereafter; RDL 50%, Development 48%, SMT 47%, Solar 48% and Welding 43%. All are LOW-confidence unit-economics assumptions. Opex/revenue declines to 34%/25%/20%/18% in 2027-2030, reflecting North Star and scale. The 2026 H2 opex bridge is solved explicitly to the upper guided full-year EBIT. Quarter weights of 15%/20%/28%/37% reflect a back-loaded equipment acceptance case, not booked orders.

3. Analytical score

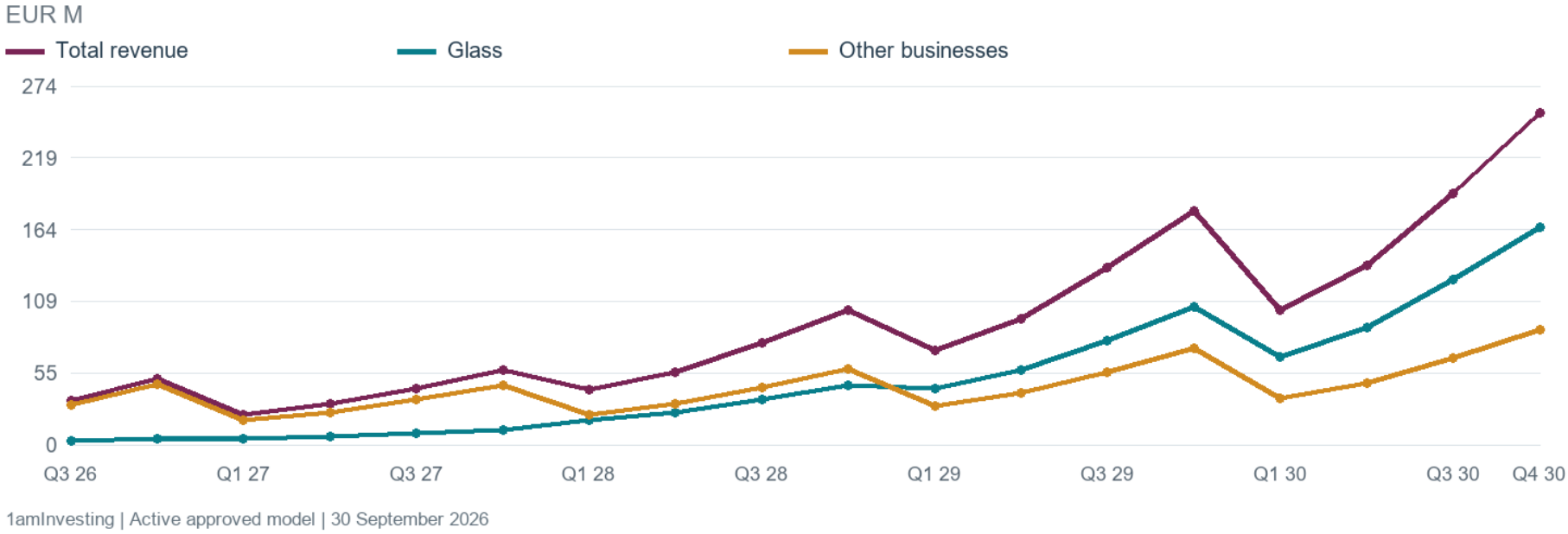
The category arithmetic gives **47/60 = 78/100** after the existing -4-point risk deduction. The portfolio Base remains **78/100**. Risk-class detail appears in section 7.

Category	Points	Maximum	Basis
Core thesis	18	20	Patent-backed LIDE process position with broad customer qualification; multi-sourcing and competing tools remain.
Cycle stage	8	10	Positioning in 2026, first ramps in 2027 and volume phase in 2029-2030.
TAM expansion	10	10	Glass packaging, photonics and perovskite/robotics adjacency extend the technology beyond one AI product.
Catalysts	5	5	A fully qualified customer has signed a production-equipment LOI; H1 2027 recognition and H2 follow-on opportunities are dated broker expectations. Conversion and acceptance remain unconfirmed. [LPK-7, LPK-8]
Macro	7	10	AI packaging demand is strong but core German industrial and solar equipment demand remains weak.
Alpha	3	5	TAM stock-versus-flow and different CPO architectures create a concrete interpretation gap.
Risks	-4	0	Four classes: competition/IP; qualification and recognition; restructuring/core demand; financing and dilution.

Product economics and quarterly operating model

The annual equipment pool is converted to incremental installed need, then product wallet and supplier capture. The approved joint scenario raises capture while delaying part of 2027 recognition. Product gross margins, operating-cost ratios and non-glass businesses are unchanged. The customer LOI and 5-8-tool report may describe the same business and are counted once.

Quarterly revenue and mix



Active current quarterly model. Source: LPK_DE_Q_Model. Values and units are shown in the table below.

Quarter	Glass	Other businesses	Total revenue
Q3 2026	2.800	30.600	33.400
Q4 2026	4.200	45.900	50.100
Q1 2027	4.639	18.456	23.095
Q2 2027	6.185	24.608	30.793
Q3 2027	8.659	34.451	43.110
Q4 2027	11.442	45.525	56.967
Q1 2028	18.381	23.286	41.667
Q2 2028	24.509	31.048	55.557
Q3 2028	34.312	43.467	77.779

Quarter	Glass	Other businesses	Total revenue
Q4 2028	45.341	57.439	102.780
Q1 2029	42.528	29.722	72.249
Q2 2029	56.704	39.629	96.333
Q3 2029	79.385	55.480	134.866
Q4 2029	104.902	73.313	178.215
Q1 2030	67.314	35.420	102.735
Q2 2030	89.752	47.227	136.980
Q3 2030	125.653	66.118	191.771
Q4 2030	166.042	87.370	253.412

Product revenue, EUR M

Quarter	Glass	RDL	Development	SMT	Solar	Welding
Q3 2026	2.800	0.400	6.760	12.000	5.440	6.000
Q4 2026	4.200	0.600	10.140	18.000	8.160	9.000
Q1 2027	4.639	0.306	4.500	6.000	3.750	3.900
Q2 2027	6.185	0.408	6.000	8.000	5.000	5.200
Q3 2027	8.659	0.571	8.400	11.200	7.000	7.280
Q4 2027	11.442	0.755	11.100	14.800	9.250	9.620
Q1 2028	18.381	1.086	4.800	6.300	6.750	4.350
Q2 2028	24.509	1.448	6.400	8.400	9.000	5.800
Q3 2028	34.312	2.027	8.960	11.760	12.600	8.120
Q4 2028	45.341	2.679	11.840	15.540	16.650	10.730
Q1 2029	42.528	2.572	5.250	6.600	10.500	4.800
Q2 2029	56.704	3.429	7.000	8.800	14.000	6.400
Q3 2029	79.385	4.800	9.800	12.320	19.600	8.960
Q4 2029	104.902	6.343	12.950	16.280	25.900	11.840
Q1 2030	67.314	4.070	5.700	6.900	13.500	5.250
Q2 2030	89.752	5.427	7.600	9.200	18.000	7.000
Q3 2030	125.653	7.598	10.640	12.880	25.200	9.800
Q4 2030	166.042	10.040	14.060	17.020	33.300	12.950

Product gross margins

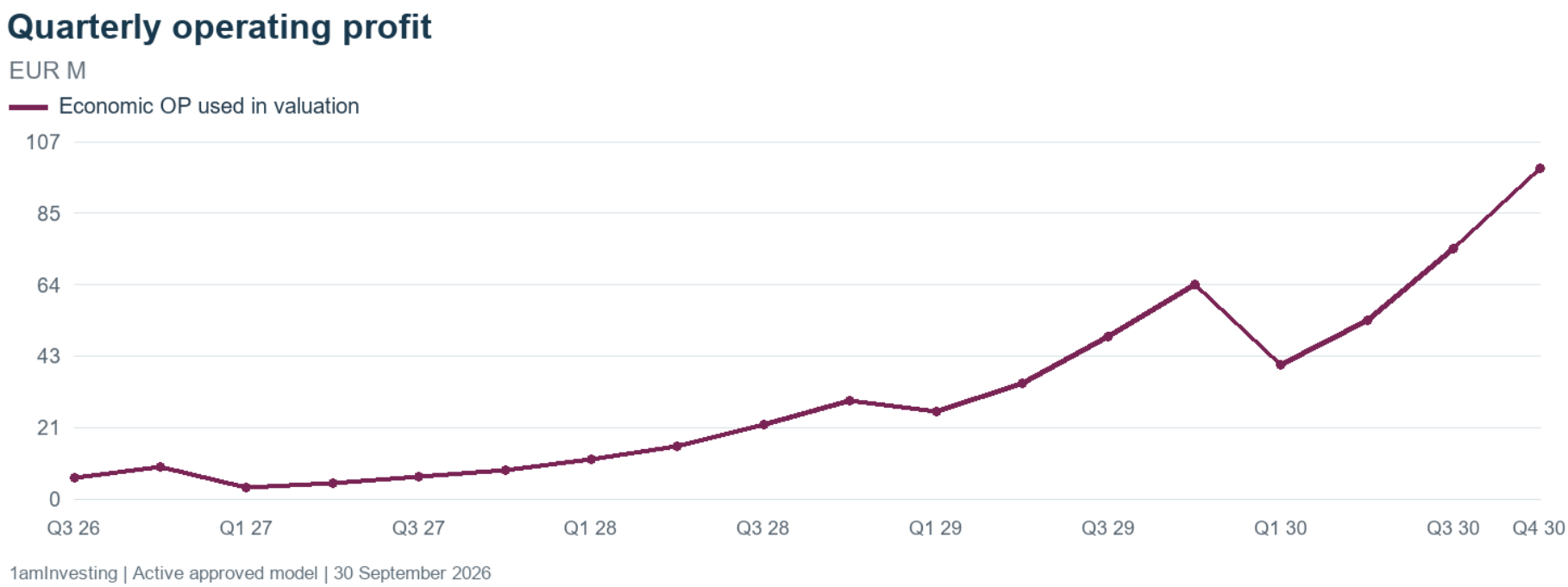
Quarter	Glass	RDL	Development	SMT	Solar	Welding
Q3 2026	55.0%	45.0%	45.0%	45.0%	45.0%	40.3%
Q4 2026	55.0%	45.0%	45.0%	45.0%	45.0%	40.3%
Q1 2027	60.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q2 2027	60.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q3 2027	60.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q4 2027	60.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q1 2028	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q2 2028	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q3 2028	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q4 2028	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q1 2029	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q2 2029	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q3 2029	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q4 2029	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q1 2030	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q2 2030	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q3 2030	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%
Q4 2030	62.0%	50.0%	48.0%	47.0%	48.0%	43.0%

Equipment stock-to-sales reference inputs

Annual assumptions below feed the reference glass-tool schedule; the active glass revenue in the product table includes the explicitly weighted capture/timing adjustment.

Year	100% tool pool EUR M	Glass adoption	Installed need	Net new need	Glass wallet	Reference capture
2026	700.000	3.0%	21.000	21.000	90.0%	70.0%
2027	900.000	8.0%	72.000	51.000	90.0%	70.0%
2028	1,150.000	22.0%	253.000	181.000	90.0%	70.0%
2029	1,420.000	48.0%	681.600	428.600	90.0%	70.0%
2030	1,700.000	80.0%	1,360.000	678.400	90.0%	70.0%

Operating profit and cost bridge



Current operating-profit path. Consolidated and parent economic profit are distinguished where minority claims apply.

Quarter	Status	Revenue	Gross profit	Opex	Consolidated OP	Valuation OP	OP / revenue
Q3 2026	Model	33.400	15.030	8.710	6.320	6.320	18.9%
Q4 2026	Model	50.100	22.545	13.065	9.480	9.480	18.9%
Q1 2027	Model	23.095	11.393	7.852	3.541	3.541	15.3%
Q2 2027	Model	30.793	15.191	10.470	4.721	4.721	15.3%
Q3 2027	Model	43.110	21.267	14.657	6.610	6.610	15.3%
Q4 2027	Model	56.967	28.103	19.369	8.735	8.735	15.3%
Q1 2028	Model	41.667	22.315	10.417	11.898	11.898	28.6%
Q2 2028	Model	55.557	29.753	13.889	15.864	15.864	28.6%
Q3 2028	Model	77.779	41.655	19.445	22.210	22.210	28.6%
Q4 2028	Model	102.780	55.044	25.695	29.349	29.349	28.6%
Q1 2029	Model	72.249	40.379	14.450	25.929	25.929	35.9%
Q2 2029	Model	96.333	53.839	19.267	34.572	34.572	35.9%
Q3 2029	Model	134.866	75.374	26.973	48.401	48.401	35.9%
Q4 2029	Model	178.215	99.602	35.643	63.959	63.959	35.9%
Q1 2030	Model	102.735	58.487	18.492	39.994	39.994	38.9%
Q2 2030	Model	136.980	77.982	24.656	53.326	53.326	38.9%
Q3 2030	Model	191.771	109.175	34.519	74.656	74.656	38.9%
Q4 2030	Model	253.412	144.267	45.614	98.653	98.653	38.9%

Annual cross-check and demand scale

Calendar buckets sum the stated quarter labels; fiscal proxy mappings remain as in the company earnings section. These totals do not replace rolling valuation windows. 2026 is only H2 in this table.

Year	Revenue EUR M	Valuation OP EUR M	OP margin
2026 H2	83.500	15.800	18.9%
2027	153.965	23.607	15.3%

Year	Revenue EUR M	Valuation OP EUR M	OP margin
2028	277.783	79.321	28.6%
2029	481.663	172.861	35.9%
2030	684.898	266.629	38.9%

Operating driver and probability transmission



Difference attributable solely to the approved breadcrumb inputs. Negative values are retained; this is not an additional revenue pool.

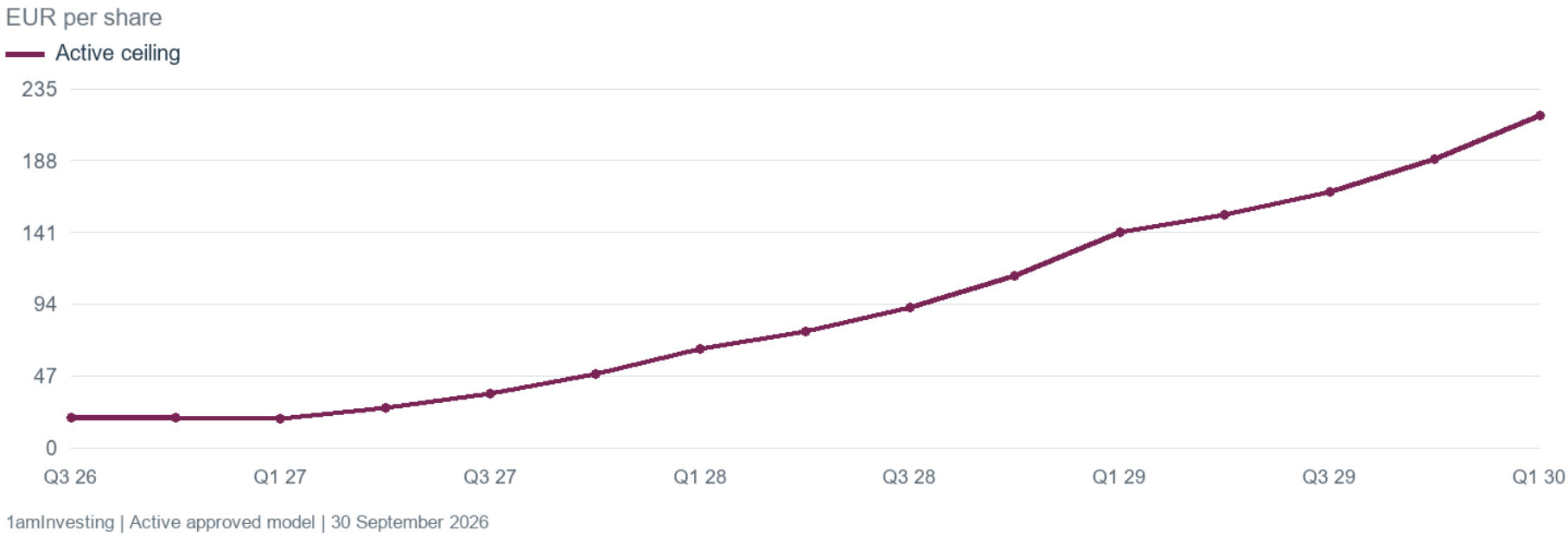
Quarter	Revenue increment EUR M	OP increment EUR M
Q3 2026	0.0000	0.0000
Q4 2026	0.0000	0.0000
Q1 2027	-0.1807	-0.0470

Quarter	Revenue increment EUR M	OP increment EUR M
Q2 2027	-0.2410	-0.0627
Q3 2027	-0.3374	-0.0877
Q4 2027	-0.4458	-0.1159
Q1 2028	1.2769	0.4725
Q2 2028	1.7026	0.6300
Q3 2028	2.3836	0.8819
Q4 2028	3.1498	1.1654
Q1 2029	2.0251	0.8506
Q2 2029	2.7002	1.1341
Q3 2029	3.7803	1.5877
Q4 2029	4.9953	2.0980
Q1 2030	3.2054	1.4104
Q2 2030	4.2739	1.8805
Q3 2030	5.9835	2.6327
Q4 2030	7.9068	3.4790

Rolling forward-four-quarter ceiling

The ceiling uses operating profit in the stated quarter plus the next three quarters. The company-specific method remains **20x operating profit**. EUR millions / 24.5M diluted shares. Same adjusted operating-profit bridge; no hypothetical funding dilution added. Probability is applied to the specified operating increment once; the ceiling receives no second probability or Base multiplier.

Rolling four-quarter ceiling



Current ceiling path. A negative-profit window is marked neg rather than assigned a positive operating-profit ceiling.

Quarter	Exact earnings window	Forward 4Q OP	Multiple	Shares millions	Unit / FX	Ceiling EUR
Q3 2026	Q3 2026 to Q2 2027	24.062	20x	24.500000	1.000	19.6428
Q4 2026	Q4 2026 to Q3 2027	24.352	20x	24.500000	1.000	19.8795
Q1 2027	Q1 2027 to Q4 2027	23.607	20x	24.500000	1.000	19.2710
Q2 2027	Q2 2027 to Q1 2028	31.964	20x	24.500000	1.000	26.0931
Q3 2027	Q3 2027 to Q2 2028	43.107	20x	24.500000	1.000	35.1892
Q4 2027	Q4 2027 to Q3 2028	58.707	20x	24.500000	1.000	47.9238
Q1 2028	Q1 2028 to Q4 2028	79.321	20x	24.500000	1.000	64.7517
Q2 2028	Q2 2028 to Q1 2029	93.352	20x	24.500000	1.000	76.2056
Q3 2028	Q3 2028 to Q2 2029	112.060	20x	24.500000	1.000	91.4775

Quarter	Exact earnings window	Forward 4Q OP	Multiple	Shares millions	Unit / FX	Ceiling EUR
Q4 2028	Q4 2028 to Q3 2029	138.251	20x	24.500000	1.000	112.8582
Q1 2029	Q1 2029 to Q4 2029	172.861	20x	24.500000	1.000	141.1111
Q2 2029	Q2 2029 to Q1 2030	186.926	20x	24.500000	1.000	152.5929
Q3 2029	Q3 2029 to Q2 2030	205.680	20x	24.500000	1.000	167.9018
Q4 2029	Q4 2029 to Q3 2030	231.935	20x	24.500000	1.000	189.3344
Q1 2030	Q1 2030 to Q4 2030	266.629	20x	24.500000	1.000	217.6560

Arithmetic and denominator sensitivity

- Q3 2026: (6.3200 + 9.4800 + 3.5410 + 4.7214) x 20 x 1 / 24.500000 = 19.6428 EUR/share.
- Q4 2026: (9.4800 + 3.5410 + 4.7214 + 6.6099) x 20 x 1 / 24.500000 = 19.8795 EUR/share.
- Q1 2027: (3.5410 + 4.7214 + 6.6099 + 8.7346) x 20 x 1 / 24.500000 = 19.2710 EUR/share.
- Q2 2027: (4.7214 + 6.6099 + 8.7346 + 11.8981) x 20 x 1 / 24.500000 = 26.0931 EUR/share.
- Q3 2027: (6.6099 + 8.7346 + 11.8981 + 15.8642) x 20 x 1 / 24.500000 = 35.1892 EUR/share.
- Q4 2027: (8.7346 + 11.8981 + 15.8642 + 22.2098) x 20 x 1 / 24.500000 = 47.9238 EUR/share.
- Q1 2028: (11.8981 + 15.8642 + 22.2098 + 29.3487) x 20 x 1 / 24.500000 = 64.7517 EUR/share.
- Q2 2028: (15.8642 + 22.2098 + 29.3487 + 25.9292) x 20 x 1 / 24.500000 = 76.2056 EUR/share.
- Q3 2028: (22.2098 + 29.3487 + 25.9292 + 34.5722) x 20 x 1 / 24.500000 = 91.4775 EUR/share.
- Q4 2028: (29.3487 + 25.9292 + 34.5722 + 48.4011) x 20 x 1 / 24.500000 = 112.8582 EUR/share.
- Q1 2029: (25.9292 + 34.5722 + 48.4011 + 63.9586) x 20 x 1 / 24.500000 = 141.1111 EUR/share.
- Q2 2029: (34.5722 + 48.4011 + 63.9586 + 39.9943) x 20 x 1 / 24.500000 = 152.5929 EUR/share.
- Q3 2029: (48.4011 + 63.9586 + 39.9943 + 53.3257) x 20 x 1 / 24.500000 = 167.9018 EUR/share.
- Q4 2029: (63.9586 + 39.9943 + 53.3257 + 74.6560) x 20 x 1 / 24.500000 = 189.3344 EUR/share.
- Q1 2030: (39.9943 + 53.3257 + 74.6560 + 98.6526) x 20 x 1 / 24.500000 = 217.6560 EUR/share.

Q3 2027 denominator case	Shares millions	Ceiling EUR
10% fewer shares	22.050000	39.0991
Active schedule	24.500000	35.1892

Q3 2027 denominator case	Shares millions	Ceiling EUR
10% more shares	26.950000	31.9902

These are arithmetic sensitivities with unchanged profit, not extra operating scenarios or forecasts of financing.

6. Industry relationships and read-throughs

TAM expansion and cross-book learning

Today LIDE enables advanced semiconductor substrates; adjacent opportunities include laser processing for robotics, medical devices and perovskite solar. Optical communication inside glass packages could create further process steps, but its architecture and adoption remain uncertain. Keep broad technology exposure in Base while demanding a distinct revenue bridge for each vector.

Apply the equipment-pool versus annual-revenue distinction to glass-substrate suppliers and equipment vendors in later batches. Within this batch, Samsung's packaging capex may validate industry demand, but it is not automatically an LPKF purchase order. Preserve different CPO meanings across LPKF, AAOI and ShunSin.

A substrate TAM measures substrate sales; an equipment pool measures installed tools; recognized equipment revenue measures accepted deliveries. Keep all three units distinct. LPKF, SCHMID, Atotech and Soitec can benefit from the same program at different steps, but none inherits another supplier's revenue or market share.

7. Risks and falsification tests

The following ledger makes the existing aggregate risk deduction explicit. It does not introduce another deduction or change the analytical score.

Risk class	Existing deduction
Competition, patent scope and production share	-1
Customer qualification, category and acceptance timing	-1
Legacy demand, restructuring and operating leverage	-1
Financing and diluted share count	-1
Total	**-4**

Risks, triggers and counters

1. Competition/IP: measure wins in operational production tools; development installations alone do not prove capture.
2. Qualification/recognition: revisit the quarter allocation when initial ramp orders, acceptance conditions and customer production schedules are disclosed. A delay changes timing directly.

3. Restructuring/core demand: test the €120M/€5.4M 2026 upper endpoint against the October 29 Q3 report and solar-order conversion. Weak legacy demand can offset glass growth.
4. Financing/dilution: use 24.5M diluted shares against 24.496546M shares reported in the retained share anchor. A 10% higher denominator lowers every ceiling by 9.09%. The September 23 release explicitly identifies a potential equity raise among financing alternatives, without terms. Keep the present denominator until there is a quantified issuance or an explicitly approved assumption. [LPK-5, LPK-7]

The operating model is LOW confidence beyond guidance because customer orders, product ASPs and quarterly shipment schedules are not public. The 20x multiple is retained.

Numbered monitoring tests

#	Checkpoint	Evidence supporting the operating case	Counter-test / failure signal
1	Production allocation	Track the September LOI becoming a binding order, H1 2027 acceptance and repeat orders at distinct customers.	LOI conversion, customer overlap or delayed acceptance may limit incremental sales; competitor wins can reduce capture.
2	Glass category	Require explicit final-package GCS/TGV evidence for a TSMC-chain allocation.	Carrier-only CoPoS milestones do not bring forward GCS revenue.
3	System yield	Track customer throughput, copper fill and panel yield alongside laser specifications.	Good standalone via specifications with poor packaged yield invalidate the recognition schedule.
4	Guidance conversion	Reconcile the next report to the revenue and adjusted-EBIT guidance bridge below.	Weak legacy orders or acceptance deferrals require changes to the affected quarter.
5	Optical adjacency	Require an identified production process, paid order and non-overlapping equipment wallet.	Research demonstrations alone receive no added optical revenue.

8. Sources, evidence gaps and model snapshot

The July earnings anchors and September evidence have distinct dates and scopes. Product allocations, forward margins and adoption remain LOW-confidence model assumptions. The September production LOI strengthens commercialization evidence; it does not disclose a full customer shipment schedule or justify a second revenue layer.

- **LPK-1** · 2026-07-23 · Corrected company H1 results. Latest actuals and full-year guidance.
- **LPK-2** · 2026-07-23 · H1 management transcript. Management speech: meaning of TAM, market share ambition, timing and capacity.

- **LPK-3** · 2026-07-23 · H1 company presentation. Roadmap and equipment-market scope.
- **LPK-4** · 2026-07-23 · H1 financial report. Segment calibration and expense basis.
- **LPK-5** · accessed 2026-09-06 · Company share information in German. 24,496,546 shares; retained 24.5M modeling denominator.
- **LPK-6** · 2026-09-01 · User-supplied workbook - User upload: v3_2_master_portfolio(1).xlsx. Uploaded numerical model and sanctioned 20x multiple.
- **LPK-7** · 2026-09-23 · LPKF production-ramp LOI announcement. Company disclosure: qualified customer, LOI value range and financing alternatives; no named buyer or fixed recognition date.
- **LPK-8** · 2026-09-28 · Montega original research distributed via EQS. Broker expectations following management discussions; H1 2027 recognition and possible later orders. Not company guidance.
- **LPK-9** · 2026-09-28 · P Equity Research expert-call summary and LIWEI_TW industry-roadmap comment, supplied by the user. Attributed channel checks, not independently verified contracts. The industry acceleration comment does not establish an additional independent customer order.
- **LPK-10** · 2026-08-18; includes August 19 response · TrendForce glass-substrate report. Secondary reporting of timing uncertainty and Samsung's denial; distinguish core material from finished substrates.

The unchanged original source document is retained separately for full research provenance. It is an archive, not an additional active valuation case. Superseded score/ceiling tables are not incorporated into this active artifact.

Model snapshot: approved 35% joint breadcrumb case adopted 30 September 2026. Analytical Base 78; protected Excel Base 78. Q3 2027 ceiling EUR 35.189231; full quarterly path above.

Model reconciliation and limitations

Current tables match LPK_DE_Q_Model and the approved proposal at every available quarter. Excel Base overrides, valuation methods, share schedules and units are preserved. The next 12-16 months deserve the highest forecast weight; outer-year estimates are substantially less reliable.

Revisit probabilities and operating inputs as evidence changes. Official confirmation is not required to value a quantified breadcrumb; source uncertainty remains explicit.